SAI VISHNU VAMSI SENAGASETTY

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AI/ML Engineer and Computer Science Master's graduate with a versatile background in architecting end-to-end intelligent systems, from high-fidelity vision models to context-aware generative AI (RAG) applications. I am dedicated to optimizing model performance, enhancing user safety through rigorous alignment and auditing, and scaling AI solutions in fast-paced, stealth-mode environments.

EDUCATION

Master of Science in Computer Science, University of Houston — August 2023 - May 2025
Bachelor of Technology in Computer Science, SRM University, AP — June 2019 – May 2023

WORK EXPERIENCE

AI Software Engineer— boomerroomers LLC- Remote — January 2026 – Present

- Architected and deployed a multi-tier RAG (Retrieval-Augmented Generation) pipeline using Next.js 15.1 and React 19, enabling the chatbot to provide contextually accurate responses by retrieving real-time product documentation and user state.
- Developed a "Guided Discovery" agentic flow that dynamically adjusts question sets based on user persona, improving user onboarding completion rates by 30% through personalized interaction.
- Optimized frontend performance by implementing modular CSS-in-JS and React Server Components for the chatbot shell, maintaining a Lighthouse performance score of 90+ despite the heavy AI logic overhead.
- Established a model evaluation framework to monitor LLM hallucinations and response drift, utilizing manual feedback loops to refine prompt templates and retrieval precision.

AI Researcher Fellow — Handshake - Remote — December 2025 – Present

- Constructed complex "red-teaming" prompt structures to probe model boundaries, identifying systemic failure modes in negative constraints and spatial reasoning that automated metrics like CLIP or FID scores failed to detect.
- Evaluated LLM code-generation performance by stress-testing models on end-to-end web application tasks, including Kanban-style systems, producing high-quality ground-truth datasets to support iterative model training.
- Authored and executed edge-case prompt engineering strategies to evaluate model sensitivity to semantic shifts, contributing to the development of more robust text-to-image encoders.
- Automated the identification of model "drifts" by establishing a baseline for aesthetic distribution and logical continuity, ensuring model outputs maintained high fidelity across diverse cinematic styles.

AI Research Intern — Grandios Mobility - Remote — August 2024 – December 2024

- Developed a hands-free ride creation engine using an Open-Source Speech-to-Text (STT) pipeline, reducing Task Completion Time (TCT) by 65%.
- Engineered custom Named Entity Recognition (NER) logic using spaCy and rule-based refinement to parse unstructured transcripts. Achieved a 92% F1-score in extracting geo-coordinates and temporal constraints by building a lightweight, domain-specific dictionary
- Architected a robust Offline-First system using SQLite and TensorFlow Lite (TFLite) for on-device intent classification.
- Developed a mock Firebase-integration layer to validate data-flow integrity, ensuring that the local SQLite intents map perfectly to a NoSQL document structure.

NLP Research Assistant — University of Houston - Houston, TX — January 2024 – July 2024

- Developed logic to parse complex HTML structures, identifying and extracting navigational items, headlines, and deep-linked page content while implementing filters for uniqueness and noise reduction.
- Implemented multiprocessing using Python's Pool class and pandas for concurrent URL processing, significantly reducing total data collection time across nearly 1,000 accessible news sources.
- Conducted multidimensional analysis of news quality using Hugging Face (HF) and MTEB libraries, evaluating sentiment, word distribution, N-grams, and reading scores to differentiate AI-generated vs. human-written content.

- Produced a labeled dataset of 27k sentences and uncovered distinct stylistic clusters via **t-SNE**, revealing that deceptive articles exhibit **3× sentiment exaggeration** and overly positive tones.

PROJECT EXPERIENCE

RAG-Based GenAI System: Personal Portfolio Assistant July 2025 – August 2025

- Built a production-style **RAG system** using **LangChain** and a Python **Flask** API, grounding LLM responses in structured domain data via prompt-based retrieval logic rather than static prompting.
- Developed an **API-driven GenAI service** in Python with request validation, logging, and response sanitization, exposing a real-time inference endpoint that supports frontend integration.

Benchmarking Model Deployment: TorchServe vs TensorFlow Serving March 2025 – April 2025

- Engineered scalable **AI-as-a-Service APIs** on **Azure ML**, deploying optimized **PyTorch** and **TensorFlow** architectures via Managed Online Endpoints. Implemented **blue/green deployment strategies** to route traffic.
- Developed **production-grade monitoring and stress-testing pipelines**, utilizing custom **multi-threaded Python scripts** to validate endpoint stability at **50 req/sec**. Optimized LSTM inference execution for CPU environments, achieving **17ms latency** and **45 req/sec throughput**.

SKILLS

Languages	Python, SQL, C++, Bash, JavaScript, HTML/CSS, R, CUDA
Frameworks	PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face, spaCy, Spark, ONNX Flutter, FastAPI, Next.js, React, Flask, TFLite, Vercel AI SDK
Tools	Docker, Kubernetes, Git, Azure ML, AWS (SageMaker, S3), Firebase Airflow, Linux, Palantir Foundry, PostgreSQL, MongoDB, Redis
Concepts	RAG, NLP, LLM Fine-tuning, NER, Computer Vision, RLHF, DPO, MLOps Model Serving, A/B Testing, CI/CD, Offline-First Architecture, CRDTs
Certifications	OCI 2025 Certified Generative AI Professional, Machine Learning A-Z (Python & R), Excel Skills for Business (JPMC), Analyzing Marketing Campaigns with Pandas, Data Manipulation in SQL

PUBLICATIONS

- **Exposing Pink Slime Journalism: Linguistic Signatures and Robust Detection Against LLM-Generated Threats** Decemeber 2025
- **SOH of Lithium-ion Batteries by Data-driven Technique with Optimized GPR** April 2023
- **A Night-Patrolling Mechanism Using Deep Learning** March 2023